MILITARY SPECIFICATION

CUPOLA, PRIMARY WEAPON CONTROL: .50 CAL, XM26, OR 20MM, XM27

1. SCOPE

1.1 Scope. This specification defines two power operated cupolas, XM26 and XM27, and one manually operated cupola, XM26, and their respective weapon control systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitations for bid or requests for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS
Military
MIL-E-55301 - Electromagnetic Compatibility.

STANDARDS
Federal
FED. STD. No. 595 - Colors.

Military
MIL-STD-193 - Painting Procedures, Tactical Vehicles (Tracked and Wheeled).
MIL-STD-417 - Rubber Compositions, Vulcanized General Purpose, Solid (Symbols and Tests).
MIL-STD-642 - Identification Marking of Administrative, Combat, and Tactical Transport Vehicles.

DRAWINGS
6590078 - Tray Assembly Ammunition Box.

PUBLICATIONS
10935870 - Cupola Engineering Parts List, .50 Cal, XM26 (Manual).
10942271 - Cupola Engineering Parts List, .50 Cal, XM26 (Power).
11619005 - Cupola Engineering Parts List, 20MM, XM27 (Power).
10935870 - Master List of Supplementary Quality Assurance Provisions for Cupola, .50 Cal, XM26 (Manual).

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10942271 - Master List of Supplementary Quality Assurance Provisions for Cupola, .50 Cal, XM26 (Power).
11619005 - Master List of Supplementary Quality Assurance Provisions for Cupola, 20MM, XM27 (Power).
10935870 - Master List of Inspection Equipment, Cupola, .50 Cal, XM26 (Manual).
10942271 - Master List of Inspection Equipment, Cupola, .50 Cal, XM26 (Power).
11619005 - Master List of Inspection Equipment, Cupola, 20MM, XM27 (Power).
10954594 - Lubricating Instructions.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction cupola. When specified (see 6.2), a preproduction cupola(s) representative of the supply contract for cupolas to be manufactured to the provisions of this specification shall be produced prior to the production manufacture of cupolas. The cupola(s) shall be subjected to Government examinations and tests to determine its capacilities relative to the requirements of this specification.

3.2 First production cupola(s). The Government selected cupola representative of early production shall be subjected to Government examination and tests to determine its conformance to the requirements of this specification.

3.2.1 Workmanship and reliability. The workmanship shall be of a quality to assure freedom from any defects that might compromise, limit, or reduce the cupola or component capabilities herein specified. Cupolas made in accordance with this specification shall be subjected to examinations and tests to prove cupola capabilities and performance.

3.2.1.1 Reliability.

(a) Cupola shall require only organizational maintenance (first and second echelon) during 2000 miles of normal vehicle operation.
(b) There shall be no component replacement which requires third through fifth echelon maintenance during 4000 miles of normal vehicle operation. (Vehicle and operation to be specified in the procurement document.)

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3.2.1.2 Reliability - XM26 (Power) and XM27 (Power). The cupola system shall have no more than 20 hours of unscheduled operator or organizational maintenance within the first 400 hours of cupola operation. The cupola shall require no field maintenance during the first 400 hours of cupola operation. Cupola operation is defined as follows:

(a) System turned on.
(b) Power controls actuated 10 times per hour.
(c) 30 percent of period with cupola mounted on a vehicle and with the vehicle stationary.
(d) 70 percent of period with cupola mounted on a vehicle and with the vehicle moving on standard vehicular durability cycle.

(Vehicle and operation to be specified in the procurement document.)

3.3 Materials.

3.3.1 Qualified products. With respect to all assemblies and component parts requiring product qualification in accordance with the pertinent specifications or drawings listed on the EPL cited in 3.4, and necessary in the construction of this cupola, the contractor shall be responsible for using only these products which are listed (by part or drawing number as QPL items) on the QPL in the referenced specification or which have been approved for inclusion on such lists. Manufacturer's inspection records shall specifically list such component(s), name of supplier(s) (as listed on the QPL or in the approval letter), and number and date of the QPL from which the selection was made or date of approval letter when products are approved but not as yet listed on the QPL.

3.3.2 Ozone resistance. Unless otherwise specified on detail drawings, rubber components which are subject to tension or flexure shall be ozone resistant to the degree required by the "C_1" test requirement of MIL-STD-417.

3.4 Construction. Components, sub-assemblies, and assemblies shall be fabricated and assembled into a complete cupola in accordance with drawings listed or referred to in Engineering Parts Lists (EPL) 10935870 XM26 (Manual), 10942271 XM26 (Power), or 11619005 XM27 (Power). All parts, sub-assemblies and assemblies requiring piece marking shall be marked as specified on drawings.

3.4.1 Hatch seals. When the hatch is closed and locked, the seals shall restrict the entrance of water to the inside of the cupola to an extent no greater than specified in 6.3(d).

3.4.2 Vision blocks. The sealed joints between the vision blocks and their mating receptacle shall restrict the entrance of water to the inside of the cupola to an extent no greater than specified in 6.3(d).

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3.4.3 Electrical circuits. The electrical circuits shall be capable of transmitting 28 volts d.c.

3.4.4 Slip ring circuits. The slip rings shall provide uninterrupted continuity of power and ground circuits throughout 360 degrees of cupola traverse.

3.4.5 Fasteners and locks. All fastening and locking devices shall be positive in action.

3.4.6 Welding repairs. Welding repairs shall be made only to the extent authorized by the procuring activity.

3.4.7 Armament mounts.

3.4.7.1 Armament mount - XM26 (Power) and XM26 (Manual). The mount assembly shall accept the .50 cal. A2 turret type machine gun and ammunition tray assembly 6590078.

3.4.7.2 Armament mount - XM27 (Power). The mount assembly shall accept the M139, 20MM automatic gun, XM120 external fire control telescope, NV2A crew served night vision device and M52 instrument light.

3.4.7.2.1 Ammunition lubricator - XM27 (Power). The ammunition lubricator shall apply lubricant to each round prior to its entry into the gun feeder.

3.4.7.2.2 Last round and emergency fire circuit - XM27 (Power). When in the normally open position, the last round switch shall deactivate the gun firing circuit. The emergency fire control switch, when placed in the "ON" position, shall override the last round circuit.

3.4.7.2.3 Gun safety lever - XM27 (Power). When placed in the "SAFE" position, the gun safety lever shall prevent the gun sear safety pin from operating. The electrical microswitch shall prevent the gun charger from operating.

3.5 Cupola performance - XM26 (Manual). The cupola completely assembled and serviced with lubricants (see 3.7) shall perform as specified herein. The cupola and fire control mechanism shall operate smoothly and uniformly in all ranges of elevation, depression, and traverse under all operating conditions. The cupola and components shall operate throughout temperature ranges between minus 65°F and plus 115°F. The cupola and mounted weapons shall operate throughout the full 360 degrees circle when on slopes up to 30 percent.

3.5.1 Weapon control system.

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3.5.1.1 Traversing mechanism - XM26 (Manual). Traversing mechanism shall traverse the cupola 360 degrees to the right and to the left. It shall incorporate a slip clutch type safety device with a non-reversible drive to prevent damage or destruction of the mechanism when the weapon barrel hits an obstruction during vehicle operation. With the cupola intalled on a vehicle at an ambient temperature of 65°F plus or minus 15°F and leveled within one degree, the average force applied tangentically to the traverse hand crank at a uniform rate to maintain cupola movement shall not exceed 22 inch-pounds in low range and 32 inch-pounds in high range with no individual reading exceeding a 26 inch-pounds in low range and 40-inch pounds in high range after breakaway.

3.5.1.2 Elevating mechanism - XM26 (Manual). The elevating mechanism shall provide the capability of vertical travel of the weapon between minus 15 degrees and plus 60 degrees. With the weapon system at an ambient temperature of 65°F plus or minus 15°F and with a full supply of ammo in the ammo tray support and box, the average force applied tangentially to the elevating hand wheel at a uniform rate to maintain weapon movement shall not exceed 35 inch-pounds with no individual reading exceeding 40 inch-pounds after breakaway. It shall incorporporate a slip clutch type safety device with a non-reversible drive (no back device) to prevent damage or destruction of the mechanism when the weapon barrel hits an obstruction during vehicle operation.

3.5.1.3 System backlash - XM26 (manual). With traverse mechanism brake applied the control system backlash shall not exceed two mils in traverse and one mil in elevation when 1000 inch-pounds are applied to the cradle in the plane of rotation, forward of the gun trunion centerline. Backlash shall be the measured angular movement of the cradle in elevation and traverse when the prescribed torque is reversed.

3.6 Performance - XM26 (Power) and XM27 (Power). Unless otherwise specified, the cupola and gun control system, completely assembled and serviced with lubricants (see 3.7), shall operate on 18 to 30 volts supplied, and meet all requirements specified herein under the following test conditions:

(a) Cupola level within one degree.
(b) System supply voltage 27.5 volts dc plus or minus 1/2 volt, unless otherwise specified (measured at slip ring outlet).
(c) Cupola combat or simulated loaded with single hatch fully open.
(d) Cupola shall traverse 360 degrees clockwise and counter-clockwise from any starting position.
(e) The cupola and control system shall be operable over a temperature range of minus 65°F to plus 125°F.

3.6.1 Gun winterization kit - XM27 (Power). When specified and installed (see 6.2), the winterization kit shall assure satisfactory gun operation in an ambient temperature of minus 25°F to minus 65°F.

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3.6.2 Electrical control box - XM26 (Power) and XM27 (Power). With the "ON or OFF" master switch of the control box in the "ON" position, the cupola electrical system supply voltage shall operate the cupola and associated controls in their respective functions. The indicator light shall show the readiness of the selected function.

3.6.2.1 Gun firing circuits - XM26 (Power) and XM27 (Power). With the cupola power switch in the "ON" position, the gun firing circuits shall operate the gun firing solenoid in the mode selected by actuating the firing switches in the cupola control handles, throghout 360 degrees rotation of cupola traverse and within the required limits of elevation and depression.

3.6.2.1.1 Gun charger circuit - XM27 (Power). When the gun charger electrical switch is actuated, the charger shall move the breech bolt assembly into the rearward (open) position. When operated manually and electrically, the charger pawl shall hold the breech bolt assembly in any position from fully closed to rearward. The warning light shall indicate if the charger pawl is not in the forward or "SAFE" position.

3.6.3 Gun charger device - XM27 (Power). The gun charger device shall charge the gun by moving the bolt into the seared position electrically and manually and shall be capable of holding the bolt in any position from fully closed to seared. Manual operation shall be accomplished by rotating the external hex head shaft with an appropriate hand wrench.

3.6.4 Cupola and gun control system.

3.6.4.1 Manual response - XM26 (Power) and XM27 (Power). With the weapon positioned initially between 0 and plus 10 degrees elevation, and the system accumulator pressure depleted, and with the control handles displaced to the maximum for elevation, 10 full strokes of the hand pump shall elevate the weapon an additional 34 degrees minimum. (This is a double acting pump: one excursion of the handle in either direction is 1 stroke.)

3.6.4.2 System backlash - XM26 (Power) and XM27 (Power). With the elevation and traverse power switch on or off, control system pressurized, and with traverse mechanism brake applied, the control systems backlash shall not exceed 1 mil total in traverse and 1 mil total in elevation when 572 inch-pounds are applied to the cradle in the plane of rotation, forward of the gun trunnion centerline. Backlash shall be the measured angular movement of the cradle in elevation and traverse when the prescribed torque is reversed.

3.6.5 Power controls - XM26 (Power) and XM27 (Power). The traverse and elevation controls and gun firing switches shall be capable of simultaneous use.

3.6.6 Cupola and gun control system deadspot - XM26 (Power) and XM27 (Power). The cupola and gun control system deadspot as measured at the power controllers shall not exceed 2.5 degrees handle movement from the neutral center position in any direction to initiate gun and cupola movement. The neutral center position

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shall be defined as that handle position which divides the initial free handle movement (backlash) in half. The deadspot angles on each side of the neutral center in the traverse axis shall be equal within one degree, and the deadspot angles in the elevation axis shall be equal within one degree. When control handle is in neutral or center position, there shall be no movement of the gun or cupola caused by turning elevation and traverse power mode switches on or off.

3.6.7 Gun elevation speeds - XM26 (Power) and XM27 (Power). The cupola power control system shall be capable of elevating and depressing the gun between minus 12.7 degrees and plus 60 degrees within the speed range of one mil per second and 87 degrees per second. With minimum supply voltage of 18 volts, gun elevation speeds shall be at least 70 degrees per second. The above requirements are applicable when starting with a fully charged accumulator.

3.6.8 Cupola traversing speeds - XM26 (Power) and XM27 (Power). The cupola power control system in low speed shall rotate the cupola in either direction at 1 mil per second, and in high speed range 1 revolution in 4.2 seconds for an average speed of 87 degrees per second for 360 degrees of rotation when measured from a standing start. The range of tracking speeds shall be from one mil per second to 60 degrees per second. With minimum supply voltage of 18 volts, cupola traverse speed shall be at least 70 degrees per second. The above require ments are applicable when starting with a fully charged accumulator. For speeds at various voltages, see Fig. 1.

3.6.8.1 Cupola traverse safety device - XM26 (Power) and XM27 (Power). At maximum traverse speed, the traverse safety device shall stop the cupola with no more than 19 degrees overtravel after the interrupter valve is actuated. These values apply with the cupola level and on a 30 percent slope.

3.6.9 Traverse, elevation and depression stability - XM26 (Power) and XM27 (Power). With cupola tilted on a 30 percent slope, the elevation and traverse power mode switches on, and the brake handle in the power position, the gun shall not move more than 2.0 mils in traverse, elevation, and depression, from a fixed position (point of aim) during a one-hour period. When using periods of less than one hour, allowable gun movement shall average not more than 1/3 mil for every 10 minutes. The above applies in power mode after hydraulic system has been operated continuously for at least 2 minutes.

3.6.10 Gun laying on moving target and tracking accuracy - XM26 (Power) and XM27 (Power). Starting with the cupola in a stationary position, the time required to lay the gun sight within the borders of a moving target from an established lead angle shall not exceed 3.5 seconds in any instance as specified in Table I. The target shall be square as specified in the table and shall be moved over a horizontal crossing course. The lead angle shall be established with the line of sight aimed behind the moving target, at which time the line of sight may be stationary or moving at a speed equal to or less than the target speed. After the gun sight has been positioned within the borders of the target, the gun sight shall remain within the target borders for the length of course in accordance with percentage of time on target specified in the table.

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Table I. Gun laying on moving target and tracking accuracy

% of Times on target	Traverse rate mils/sec (ref.)	Established lead mils	Target size mils (square)	Target size inches (square)	Target range meters	Target velocity mph	Length of course meters
100	1.0	2.0	1.25	90.0	2000	4.0	90
100	3.3	7.5	1.25	90.0	2000	13.5	185
100	9.8	20.0	1.25	90.0	2000	40.0	185
100	19.6	7.5	1.25	13.5	275	12.0	365
100	32.6	10.0	1.25	13.5	275	20.0	365
100	65.2	20.0	5.00	54.0	275	40.0	365

3.6.11 Gun laying on stationary target - XM26 (Power) and XM27 (Power). From a layoff position tabulated in Table II, the average time required for a total of 4, 8 or 12 trials to position the gunsight reticle within the borders of a 0.25 mil square target shall not exceed that tabulated in the table. The time for layoff position number 3 shall be met for an equal number of trails from each of 4 different quadrants. The time of all other layoff positions shall be met for two directions. The results of any 1 series of trials shall not be used in any other series.

Table II. Gun laying on stationary target

Position	Layoff - mils		Average Time - Seconds	
	Azimuth	Elevation depression	Power Controls	Manual Controls
1	--	∠ 10	2	2
2	25	--	3	3
3	25	∠ 10	5	5
4	100	--	4	11.5
5	400	--	5	13.5
6	800	--	6	
7	1600	--	8	
8	3200	--	13	

3.6.12 Hydraulic pressure limit switch - XM26 (Power) and XM27 (Power). When power traversing the cupola, the hydraulic pressure switch shall activate the system pump electric motor at a control system pressure between 1525 psi and 1600 psi with the power controls in the neutral position, the hydraulic pressure switch shall deactivate the system pump motor at a control system pressure between 1800 psi and 1900 psi.

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3.6.13 Cupola control system fluid leakage - XM26 (Power) and XM27 (Power). During operation of the cupola gun control system, there shall be no evidence of fluid leakage to an extent greater than specified in 6.3(c). When the system is not in operation there shall be no evidence of leakage to an extent greater than specified in 6.3(b). Hydraulic lines shall not leak.

3.6.14 Gun fire control system. Target aiming point is at 1000 meters. For boresighting at distances between 40 feet and 1000 meters, use dimensions, shown on Figure 2 as a basis for calculations.

3.6.14.1 Boresight adjustment of external sights. Lateral and vertical boresight adjustment ranges shall be no less than plus or minus 5 mils of target aiming point.

3.6.14.2 Boresight adjustment of direct fire telescope mount - XM27 (Power. When the direct fire telescope is specified, lateral and vertical boresight adjustment ranges shall be no less than plus or minus 5 mils of the target aiming point.

3.6.14.3 Boresight adjustment of night vision device - XM27 (Power). When the crew served night vision device is specified, lateral and vertical boresight adjustment ranges shall be no less than plus or minus 5 mils of the target aiming point.

3.7 Lubrication. Cupola shall be lubricated in accordance with lubricating instruction 10954594.

3.8 Electromagnetic compatibility. Complete cupola shall be radio interference suppressed in accordance with requirements of MIL-E-55301.

3.9 Painting and marking.

3.9.1 Painting. Unless otherwise specified, the exterior and interior of cupola component assemblies and parts, which require painting, shall be prepared and painted in accordance with applicable drawings and MIL-STD-193. Cupola exterior paint shall match color chip X24087 (olive drab) of Federal Std. No. 595.

3.9.2 Marking. Cupola marking shall be in accordance with MIL-STD-642.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract, or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein and may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Suppliers quality control system. Unless otherwise specified by the procuring activity, the supplier shall provide and maintain an inspection and quality control system acceptable to the Government. A written description of this system shall be submitted to the Contracting Officer prior to initiation of production. The supplier will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality control procedure. The supplier shall obtain Government approval for any change to the written procedure that might affect the degree of control required by this specification or other applicable documents referenced herein.

4.1.2 Parts, components and assemblies. Parts, components and assemblies shall be inspected in accordance with the drawings listed or referenced on EPL 10935870, EPL 10942271 or EPL 11619005, the Master List of Supplementary Quality Assurance Provisions 10935870, 10942271 or 11619005, and the Master List of Inspection Equipment Drawings 10935870, 10942271 or 11619005.

4.1.3 Government verification. All quality assurance operations performed by the supplier are subject to Government verification at unscheduled intervals. Verification shall consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to determine quality of product being offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an effect upon the quality of the product, shall immediately be called to the attention of the supplier. Failure of the supplier to promptly correct malpractices may be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.4 Inspection equipment drawings. Inspection equipment drawings listed on Master List of Inpsection Equipment Drawings 10935870, 10942271 or 11619005 will serve as stantard for determining the adequacy of supplier's inspection equipment (see 6.4).

4.1.5 Qualified products. The supplier's inspection records shall be checked to determine supplier conformance to 3.3.1.

4.2 Preproduction and first production inspection.

4.2.1 Preproduction cupola inspection . A preproduction cupola (see 3.1), when furnished, shall be inspected at a location approved by the Government, to ascertain conformance with the requirements of this specification. Failure of the preproduction cupola to comply with any of the requirements specified may be cause for refusal by the Government to conduct a retest until acceptable corrective measures have been taken.

4.2.2 First production cupola inspection.

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4.2.2.1 Special process inspection. An in-process inspection shall be conducted by the Government representatives during fabrication of the first production cupola (see 3.2), to evaluate conformance to materials and workmanship with drawing and specification requirements.

4.2.2.1.1 Cupola inspection. The cupola shall be inspected prior to covering the weldments with paint or other pertinent components or assemblies.

4.2.2.2 Complete cupola inspection. Subsequent to the supplier's inspection of the first completed cupola, the Government shall inspect this cupola at the supplier's plant to determine conformance to contract and specification requirements.

4.2.2.2.1 Corrections. Corrections found necessary as a result of above inspection shall be made by the supplier.

4.2.2.3 Production cupola tests. After completion of the cupola inspection, a second cupola (one of the first ten) selected from the first month's production shall be subjected to examination and tests as specified in 4.4.1. Subsequent to examination and tests, the selected cupola shall be subjected by the Government to tests specified in Table III and 4000 miles on-vehicle testing as specified in the procurement document at a site approved by the Government, which will require a maximum of 90 days. Delays caused by cupola failure due to poor quality of workmanship or material, or failure of the supplier to comply with specification or drawing requirements shall not be the basis for adjustment of the contract performance dates or delivery schedule or contract price. Under separate contractual authority, the supplier will expeditiously furnished cupola repair parts, as required to adequately support the above tests. Delays caused by cupola failure and failure of the supplier to adequately support the cupola with parts during the tests shall not be the responsibility of the Government.

Table III. 400 hour test (Government Proving Ground)

Vehicle Condition	Percent of Time
Stationary	30
Moving	70

NOTE: Cupola installed on a vehicle and with the power control system turned on and actuated 10 times per hour.

4.2.2.3.1 Test failure. Failure of the production cupola tested in 4.2.2.3 to meet requirements specified, due to major deficiencies of workmanship or material nature, will be cause for the Government to refuse to continue acceptance of production cupolas until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Any deficiencies found as a result of the test will be considered prima facie evidence that

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all cupolas accepted prior to the completion of the test are similarly deficient unless evidence to the contrary, satisfactory to the Contracting Officer, is furnished. Such deficiencies on all cupolas shall be corrected by the supplier at no cost to the Government.

4.3 Classification of tests.

(a) Acceptance tests 4.4.1.1 through 4.4.1.10.1.
(b) Control tests 4.4.2 through 4.4.2.6.
(c) Inspection Comparison Test (ICC) 4.4.3 through 4.4.3.6.

4.4 Inspection provisions.

4.4.1 Cupola examination and acceptance tests. Each completed cupola shall be operated at place of manufacturer by the contractor and subjected to examination (see Table IV), and subjected to acceptance (see 4.3 a and Table V). Cupola shall be free of any defects specified in Table IV, and shall meet all performance requirements specified herein.

4.4.1.1 Test failure. If a cupola fails to pass any acceptance test specified herein, the Government shall not accept it until evidence has been provided by the supplier that corrective action has been taken.

Table IV. Examination characteristics

Cupola assembly: Structural and welding defects, malfunction, stability, burrs, sharp edges, interference, torque.
Hatch cover and vision blocks: Malfunction, holdopen and locking devices, damage, improper fit, adjustment, seal installation.
Electrical system components and circuits: Malfunction, improper assembly, connections, cables properly routed and secured.
Tubing and hoses: Improper assembly, installation, properly routed and secured, connections, damage, leaks (see 6.3).
Cupola and gun control system: Improper assembly, installation, malfunction, elevation speeds, traversing speeds and acceleration range, tracking accuracy, gun laying, response, backlash, manual and power controls, traverse safety device, damage and leaks (see 6.3).
Painting and protective coating: Application, coverage and materials improper.
Lube fittings and lubrication: Fittings missing, defective or improper installed, lubrication improper.

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Table V. Location of tests

Title	Paragraph Numbers	Place of Manufacture	Proving Ground (Government)
Acceptance test	4.4.1.1	X	X
Hatch and vision block seals leakage check	4.4.1.2	X	X
Electrical circuits check	4.4.1.3	X	X
Slip ring circuit check	4.4.1.3.1	X	X
Fasteners and lock check	4.4.1.4	X	X
Armament mount	4.4.1.5	X	X
Armament mount check	4.4.1.5.1	X	X
Ammunition lubricator check - XM27 (Power)	4.4.1.5.2	X	X
Last round and emergency fire circuit check - XM27 (Power)	4.4.1.5.3	X	X
Gun safety lever check - XM27 (Power) XM26 (Manual)	4.4.1.5.4	X	X
Cupola performance check - XM26 (Manual)	4.4.1.6	X	X
Weapon control system	4.4.1.7	X	X
Traverse mechanism check - XM26 (Manual)	4.4.1.7.1	X	X
Elevation mechanism check - XM26 (Manual)	4.4.1.7.2	X	X
Traverse system backlash check - XM26 (Manual)	4.4.1.7.3	X	X
Elevation system backlash check - XM26 (Manual)	4.4.1.7.4	X	X
Cupola performance check - XM26 (Power) - XM27 (Power)	4.4.1.8	X	X
Electric control box check - XM26 (Power) - XM27 (Power)	4.4.1.8.1	X	X
Gun firing circuits check - XM26 (Power) - XM27 (Power)	4.4.1.8.2	X	X
Gun charger device check-XM27 (Power)	4.4.1.8.3	X	X
Cupola and gun control system	4.4.1.9	X	X
Manual response check - XM26 (Power) - XM27 (Power)	4.4.1.9.1	X	X
System backlash check - XM26 (Power) - XM27 (Power)	4.4.1.9.2	X	X
Power controls check - XM26 (Power) - XM27 (Power)	4.4.1.9.3	X	X
Cupola and gun control system dead-spot check - XM26 (Power) - XM27 (Power)	4.4.1.9.4	X	X

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Table V. Location of tests (Continued)

Title	Paragraph Numbers	Place of Manufacture	Proving Ground (Government)
Gun elevation speeds check - XM26 (Power) - XM27 (Power)	4.4.1.9.5	X	X
Elevation and depression limits limits check - XM26 (Power) - XM27 (Power)	4.4.1.9.5.1	X	X
Cupola traverse speed check - XM26 (Power) - XM27 (Power)	4.4.1.9.6	X	X
Cupola traverse safety device check - XM26 (Power) - XM27 (Power)	4.4.1.9.7	X	X
Traverse, elevation and depression stability check - XM26 (Power) - XM27 (Power)	4.4.1.9.8	X	X
Hydraulic pressure limit switch check - XM26 (Power) - XM27 (Power)	4.4.1.9.9	X	X
Cupola control system fluid leakage check - XM26 (Power) - XM27 (Power)	4.4.1.9.10	X	X
Gun fire control system check	4.4.1.10	X	X
Boresight adjustment of external sight check	4.4.1.10.1	X	X
Control tests	4.4.2	X	X
Cupola traversing speed, gun laying on moving target, tracking accuracy check - XM26 (Power) - XM27 (Power)	4.4.2.3	X	X
Gun laying on stationary target check XM26 (Power) - XM27 (Power)	4.4.2.4	X	X
Boresight adjustment of direct fire telescope mount check - XM27 (Power)	4.4.2.5	X	X
Boresight of night vision device check - XM27 (Power)	4.4.2.6	X	X
Inspection comparison test	4.4.3	X	X
Environmental performance test	4.4.3.2		X
Gun winterization kit test	4.4.3.3		X
Electrical firing circuit test	4.4.3.4		X
Gun laying on moving target and tracking accuracy test - XM26 (Power) - XM27 (Power)	4.4.3.5		X
Electromagnetic compatability test	4.4.3.6		X

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4.4.1.2 Hatch and vision block seals leakage check. To determine conformance to 3.4.1 and 3.4.2, the hatch shall be closed and locked and shall not exhibit leakage as specified while a spray of free flowing water from a 1/2 to 3/4 inch inside diameter hose is directed on the hatch and vision blocks from a distance of three (3) feet for a period of three (3) minutes.

4.4.1.3 Electrical circuits check.

4.4.1.3.1 Slip ring circuit check. To determine conformance to 3.4.4, the slip ring shall be checked for continuity through a 360 degree rotation phase of operation. There shall be no evidence of discontinuity or shorting throughout this operation.

4.4.1.4 Fasteners and locks check. To determine conformance to 3.4.5, all fasteners and locks shall be functionally checked.

4.4.1.5 Armament mounts.

4.4.1.5.1 Armament mount check. To determine conformance to 3.4.7.1 and 3.4.7.2, the XM26 (Power) and XM26 (Manual) armament mount assemblies shall be tested for acceptance of the specified machine gun. The XM27 (Power) armament mount assembly shall be checked for acceptance of the 20mm gun.

4.4.1.5.2 Ammunition lubricator check - XM27 (Power). To determine conformance to 3.4.7.2.1, the lubricator cylinder shall be pressurized by operation of the handle and checked for function by depressing the plunger in the manifold assembly. Results shall be as specified.

4.4.1.5.3 Last round and emergency fire circuit check - XM27 (Power). To determine conformance to 3.4.7.2.2, the "EMER. FIRE" control switch shall be placed in the "ON" position. The indicator light shall light. Actuating the firing switches in the control handles shall cause the firing solenoid to operate with the last round switch in the normally open and closed positions. The gun firing circuit shall be inoperable with the "EMER. FIRE" control switch in "OFF" position and the last round switch in open position.

4.4.1.5.4 Gun safety lever check - XM27 (Power). To determine conformance to 3.4.7.2.3, the gun firing safety lever shall be placed in the "SAFE" position and the gun firing triggers actuated. The safety lever shall prevent operation of the firing mechanism as specified.

4.4.1.6 Cupola performance check - XM26 (Manual). To determine conformance to 3.5 for acceptance testing, the cupola shall be leveled to within one degree and the cupola and fire control mechanism functionally tested at ambient temperature for performance as specified. The cupola shall also exhibit functional capabilities when positioned on a 30 percent slope.

4.4.1.7 Weapons control system.

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4.4.1.7.1 Traversing mechanism check - XM26 (Manual). To determine conformance to 3.5.1.1, the cupola shall be operated 360 degrees clockwise and counterclockwise in low and high range by applying force tangentially to the traverse hand crank at a uniform rate to maintain cupola movement for each direction of traverse. The average force required to maintain movement, with no individual reading in excess of the specified values, shall be in compliance with requirements.

4.4.1.7.2 Elevating mechanism check - XM26 (Manual). To determine conformance to 3.5.1.2, the weapon system shall be functionally tested in elevation and depression at an ambient temperature of 65°F plus or minus 15°F with the ammo tray support and box combat or simulated loaded. The average force required to maintain movement, with no individual reading in excess of the specified value, shall be in compliance with requirements for vertical travel of weapon between minus 15 degrees depression and plus 60 degrees elevation.

4.4.1.7.3 Traversing system backlash check - XM26 (Manual). To determine conformance to 3.5.1.3, force shall be applied as required and the measured backlash shall not exceed the value specified.

4.4.1.7.4 Elevating system backlash check - XM26 (Manual). To determine conformance to 3.5.1.3, force shall be applied as required and the measured backlash shall not exceed the value specified.

4.4.1.8 Performance check - XM26 (Power) - XM27 (Power). To determine conformance to 3.6, (a) through (d), for acceptance testing, the cupola shall be leveled to within one degree and the cupola and fire control mechanism functionally tested at ambient temperature for performance with the supplied voltages as specified.

4.4.1.8.1 Electrical control box check - XM26 (Power) and XM27 (Power). To determine conformance to 3.6.2, the Master "ON/OFF" switch shall be actuated and current supply for operation determined. For the XM27 (Power), the switches shall be actuated and the indicator lights shall function as specified.

4.4.1.8.2 Gun firing circuits check - XM26 (Power) and XM27 (Power). To determine conformance to 3.6.2.1, the "FIRE" control switch shall be placed in the "ON" position and the control handles gun firing switches actuated to demonstrate operation of the firing circuit in each of the firing mode positions. Results shall be as specified.

4.4.1.8.3 Gun charger device check - XM27 (Power). To determine conformance to 3.6.2.1.1 and 3.6.3 thee charger "START" switch shall be actuated and the indicator light shall light as specified. The "NORM-OFF-JOG" switch shall be actuated

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from "OFF" to "JOG" to "OFF" positions to demonstrate capability of the mechanism to complete its charging cycle and to hold in any position as specified. With the gun charger switches in open position, the charger mechanism shall be manually operated as specified and shall demonstrate capability to charge the gun.

4.4.1.9 Cupola and gun control system.

4.4.1.9.1 Manual response check - XM26 (Power) and XM27 (Power). To determine conformance to 3.6.4.1, the gun shall be positioned within the degrees of elevation specified and the control system depeleted. Displace systems control handles to maximum position for elevation and pump manual pump handle 10 full strokes. Gun shall elevate not less than the additional degrees specified.

4.4.1.9.2 Systems backlash check - XM26 (Power) and XM27 (Power). To determine conformance to 3.6.4.2, the cupola and gun control system shall be tested for backlash in elevation and traverse by applying force as prescribed. Backlash for elevation and traverse shall be as specified.

4.4.1.9.3 Power controls check - XM26 (Power) and XM27 (Power). To determine conformance to 3.6.5, the cupola shall be traversed clockwise and counterclockwise, the gun elevated and depressed and the gun firing switches actuated at random positions. Resultant simultaneous capabilities shall be as specified.

4.4.1.9.4 Cupola and gun control system deadspot check - XM26 (Power) and XM27 (Power). To determine conformance to 3.6.6, the power control handles shall be operated as specified and the deflection of the handles to initiate movement of the cupola and gun measured. Angular displacement of the control handles to initiate movement first in one direction, then reversed from neutral center, shall be not more than specified and shall be equal as specified. With control handles in neutral position, the elevation and power mode switches shall be turned on and off not less than three times and no movement of the gun or cupola shall be observed.

4.4.1.9.5 Gun elevation speeds check - XM26 (Power) and XM27 (Power). To determine conformance to 3.6.7 for acceptance testing, the gun shall be operated in elevation and depression at a one mil per second rate for not less than 5 seconds and checked for control of speed. Maximum speed achievement for applied supply voltages shall be as specified.

4.4.1.9.5.1 Elevation and depression limits check - XM26 (Power) and XM27 (Power). To determine conformance to 3.6.7, the centerline of gun bore shall be positioned parallel to the plane of the cupola bearing assembly, and then depressed to maximum depression and elevated to maximum elevation. In each instance, the degree of travel shall be as specified.

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4.4.2.1 Frequency. The first cupola off the production line, or one of the first ten cupolas produced, and thereafter one per month, shall be checked in accordance with the following control tests.

4.4.2.2 Test failure. If a cupola fails to pass any control tests specified herein, the Government may stop acceptance of further cupolas until evidence has been provided by the supplier that corrective action has been taken.

4.4.2.3 Cupola traversing speed, gun laying on moving target, and tracking accuracy check - XM26 (Power) and XM27 (Power). To determine conformance to 3.6.8 and 3.6.10 for control tests, the tracking rate of one mil per second as shown in Table I shall be used. A suitable cupola tracking instrument shall be used as an equivalent means of making this test. All ranges and target information in Table I will be used when conducting the ICT at a Government Proving Ground.

4.4.2.4 Gun laying on stationary target check - XM26 (Power) and XM27 (Power). To determine conformance to 3.6.11, the gunsight reticle shall be brought within the border of the specified target for position 3 within the selected number of trials from four different quadrants and from two directions for positions 5 and 7, using manual controls. All positions in table II will be used when conducting the ICT at a Government Proving Ground.

4.4.2.5 Boresight adjustment of direct fire telescope mount check - XM27 (Power). To determine conformance to 3.6.14 and 3.6.14.2, when applicable, the direct fire telescope mount assembly shall be mounted and a master telescope locked in place. The line of sight shall be adjusted on its aiming point relative to the centerline of weapon tube and the lateral and vertical boresight adjustment screws rotated. Adjustment ranges shall be as specified.

4.4.2.6 Boresight adjustment of night vision device check - XM27 (Power). To determine conformance to 3.6.14 and 3.6.14.3, when applicable, the night vision device shall be mounted and adjusted on its aiming point relative to the centerline of weapon tube. Lateral and vertical boresight adjustment ranges shall be accomplished and shall be as specified.

4.4.3 Inspection comparison test. The Government may select cupolas at any time during the contract production period and subject cupolas to applicable tests specified in Table V and Table VI (Government Proving Ground Tests). Tests shall be conducted by the Government at a site selected or approved by the Government. Cupolas selected for Government tests shall not include any cupolas previously tested for conformance to 4.4.2 (control tests).

4.4.3.1 Test failure. Failure of any cupola tested to comply with any of the requirements specified herein due to major deficiency of a workmanship or materials nature occurring during or as a result of the test cycle, may be cause for the Government to refuse to continue acceptance of cupolas until evidence has been provided by the supplier that corrective action has been taken to eliminate the condition.

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Table VI. 400 hour test (Government Proving Ground)

Vehicle condition	Percent of time
Stationary	30
Moving	70

NOTE: Cupola installed on a vehicle and with the power control system turned on and actuated 10 times per hour.

4.4.3.2 Environmental performance test. To determine conformance to 3.6 (a) through (e), the selected cupola shall be operated under the specified conditions while mounted on a vehicle or test stand and checked for operational requirements as listed in Table V.

4.4.3.3 Gun winterization kit test - XM27 (Power). To determine conformance to 3.6.1, the gun winterization kit shall be installed in accordance with applicable drawings and specifications and shall be subjected to ambient temperatures of minus 25°F to minus 65°F. The gun shall function as specified.

4.4.3.4 Electrical firing circuit test - XM27 (Power). To determine conformance to 3.6.2, the cupola fire control circuits shall be tested as specified and observed for functional requirements.

4.4.3.5 Gun laying on moving target and tracking accuracy tests - XM26 (Power) and XM27 (Power). To determine conformance to 3.6.10, the cupola shall be tested as specified at traverse rates shown in Table I.

4.4.3.6 Electromagnetic compatibility. To determine conformance to 3.8, the cupola shall be subjected to a radio interference suppression test and must meet the requirements as specified in the applicable specification.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery and storage shall be in accordance with the applicable packaging data sheet.

6. NOTES

6.1 Intended use. This cupola is intended for use by the Armed Forces on combat vehicles.

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6.2 Procurement document should specify the following:

(a) Title, number and date of this specification.
(b) Preproduction cupola, if required (see 3.1).
(c) Special kits, if required (see 3.6.1).
(d) Level of preservation packaging and delivery (see 5.1).

6.3 The following shall be used with the examination characteristics for leaks:

(a) Weep: Any non-recurring evidence of fluid beyond the seal.
(b) Seep: Any recurring evidence of fluid beyond the seal that does not result in the formation of a droplet.
(c) Droplet: Any recurring evidence of fluid beyond the seal that results in formation of a non-falling droplet.
(d) Drip: Any recurring evidence of fluid beyond the seal where droplet(s) form(s) and fall(s).

6.4 Inspection equipment calibration and inspection. In inspecting the supplier's inspection equipment, the Government will determine that the supplier has available, and utilizes correctly, gaging, measuring and test equipment of the required accuracy and precision. Except as otherwise specified, the supplier will have available a set of master gages, standards and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the supplier and made available for review by the Government. The calibration of gages, standards, and instruments will be periodically checked by authorized Government personnel.

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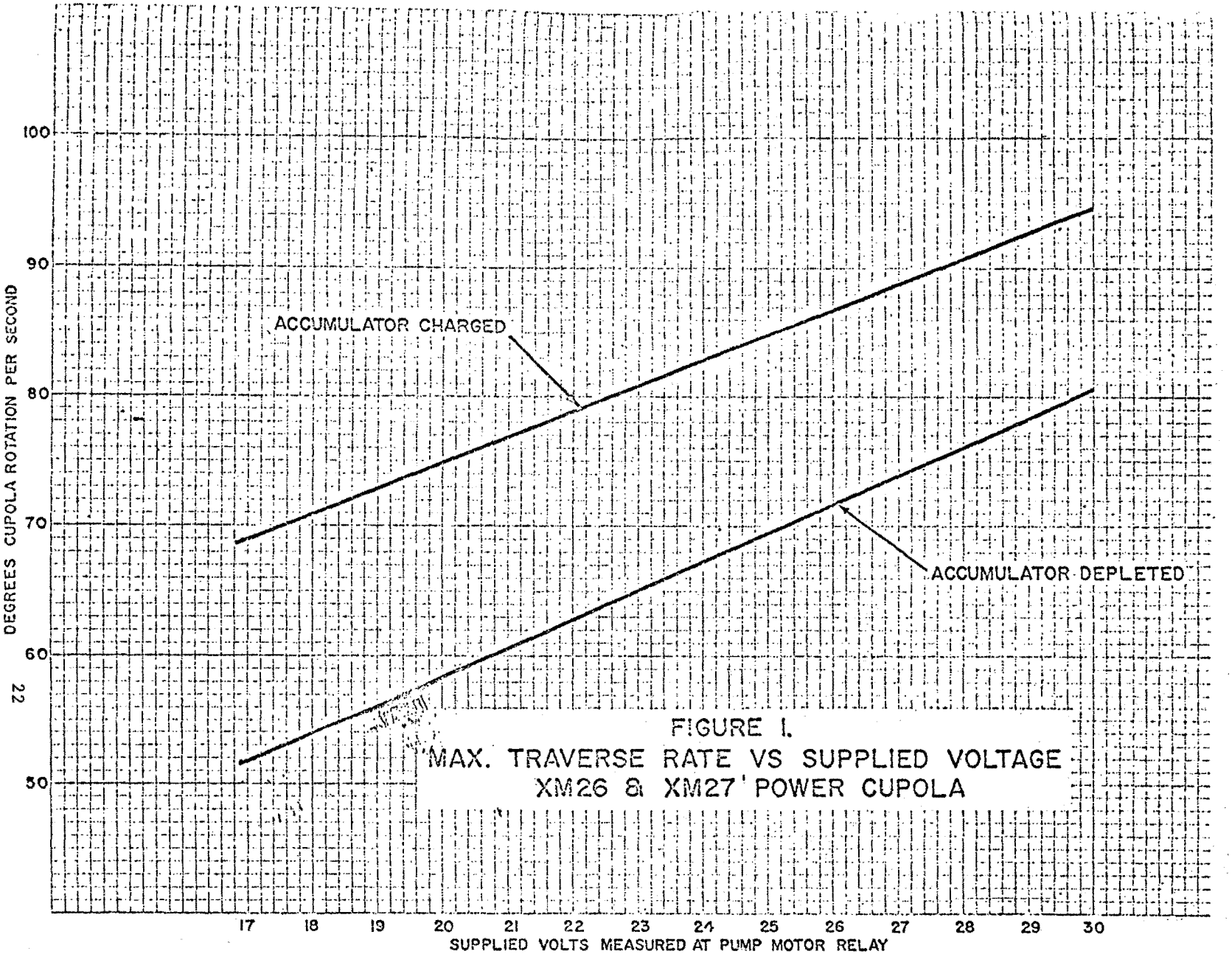
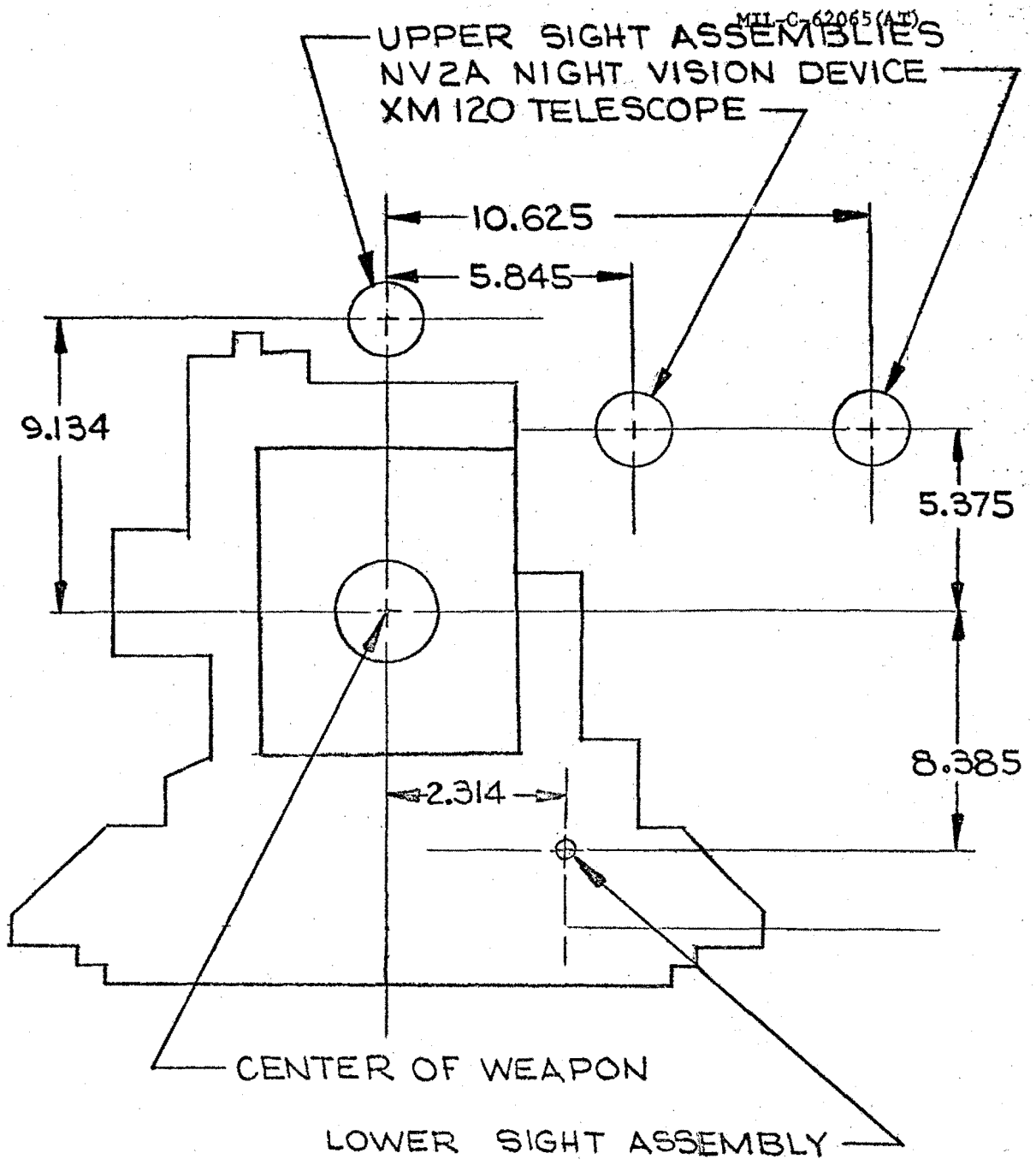


FIGURE 1.
MAX. TRAVERSE RATE VS SUPPLIED VOLTAGE
XM26 & XM27 POWER CUPOLA

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XM 26 AND XM 27 CUPOLA

FRONT VIEW

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FIGURE 2